Workshop 1 — Requirements and User Stories

Kintsugi Studio — Online Booking and Business Management Platform

Software Engineering II — 2026-II

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Date: November 2026

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1. Introduction and Business Context

Kintsugi Studio is a beauty and transformation studio located in Bogotá, Colombia. It serves approximately **200–230 clients per month** and operates from a single site with a team of professionals. Today, appointments are coordinated through WhatsApp and manual notes, while client history and financial information are kept in spreadsheets.

The studio requested an integrated platform with two faces:

- A **public website** where clients discover services and book real appointments.
- An **internal management panel** with agenda, client records, notifications, financial control, inventory, providers, and reports.

This workshop defines the functional and non-functional requirements, the user stories with acceptance criteria, the user story map, the MVP boundary, and the initial estimation.

Scope note: The client marked online payments, automatic confirmation, electronic invoicing, loyalty points, multi-site, provider purchase orders, and accounting integration as *Won't* in the MoSCoW exercise. They remain in the roadmap but are not part of the MVP.

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2. Functional Requirements

2.1 Public website

ID	Requirement
RF-01	The system shall display the studio's brand content: home, philosophy, gallery, contact information, location, social media links, and policies.
RF-02	The system shall display a service catalog with name, description, duration, price, and assigned professionals.
RF-03	The system shall allow visitors to filter services by category and professional.
RF-04	The system shall allow visitors to contact the studio through WhatsApp and a contact form.
RF-05	The system shall publish privacy, personal data treatment, cancellation, and refund policies.

2.2 Booking

ID	Requirement
RF-06	The system shall allow a client to select a service, a professional (or "any available"), and a date.
RF-07	The system shall calculate and display real-time availability based on working hours, schedule blocks, service duration, and existing appointments.
RF-08	The system shall hold the selected slot temporarily while the client completes the booking.

ID	Requirement
RF-09	The system shall prevent overlapping appointments for the same professional.
RF-10	The system shall register the appointment with status `pending` and confirm it according to the configured rule.
RF-11	The system shall allow the client to reschedule or cancel within the configured policy window.
RF-12	The system shall generate an ICS file and synchronize the appointment with Google Calendar.

2.3 Client account and communications

ID	Requirement
RF-13	The system shall allow clients to register and log in with email and password or a one-time code.
RF-14	The system shall show clients their upcoming and past appointments in read-only mode.
RF-15	The system shall record marketing consent and allow clients to withdraw it.
RF-16	The system shall support the sale and redemption of virtual gift cards.

2.4 Agenda and CRM

ID	Requirement
RF-17	The system shall provide a daily and weekly agenda view by professional.
RF-18	The system shall allow administrators to create, move, confirm, cancel, and mark no-show appointments.
RF-19	The system shall store a client record with contact data, service history, formulas, photographs, notes, allergies, and preferences.

ID	Requirement
RF-20	The system shall detect possible duplicate clients by normalized phone, email, or document number.
RF-21	The system shall enforce roles, two-factor authentication for administrators and operators, and audit every sensitive change.

2.5 Finance, inventory, and reports

ID	Requirement
RF-22	The system shall support daily cash closing with income, expenses by category, and payment method breakdown.
RF-23	The system shall calculate professional commissions (55 % / 60 %) and support weekly settlement.
RF-24	The system shall manage inventory, providers, provider invoices, and generate daily and monthly reports and dashboard indicators.

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3. Non-Functional Requirements

ID	Quality attribute	Requirement and metric
RNF-01	Usability	A new client shall complete a booking in no more than four steps. The interface shall be mobile-first and follow WCAG 2.1 AA basics.
RNF-02	Performance	The availability query shall respond in under 2 seconds at the 95th percentile. The public home page shall reach LCP under 2.5 seconds on a 4G connection.

ID	Quality attribute	Requirement and metric
RNF-03	Availability	The system shall be available 99.5 % of the month, excluding announced maintenance windows.
RNF-04	Security	All traffic shall use HTTPS. Passwords shall be hashed. Inputs shall be validated. The application shall mitigate the OWASP Top 10 risks.
RNF-05	Authentication	Administrator and operator accounts shall use TOTP two-factor authentication. Sessions shall expire after 30 minutes of inactivity, and accounts shall lock after five failed attempts.
RNF-06	Authorization	The system shall enforce role-based access and object-level permissions. Professionals shall only access their own agenda and history. Clients shall only access their own data.
RNF-07	Auditability	Every price change, cancellation, payment, refund, client edit, and permission change shall be logged with actor, timestamp, and before/after values.
RNF-08	Privacy	Personal data shall be processed under Law 1581 of 2012. Consent shall be explicit and revocable. Data shall be retained for one year unless a legal exception applies.

ID	Quality attribute	Requirement and metric
RNF-09	Reliability and backup	The system shall run daily backups with an RPO of 24 hours and an RTO of 4 hours. Restore tests shall be performed quarterly.
RNF-10	Maintainability	Domain logic shall have at least 70 % automated test coverage. Every pull request shall pass lint, type checking, and tests in CI.
RNF-11	Portability	The system shall run with Docker and environment-based configuration. No secret shall be hardcoded in the repository.
RNF-12	Concurrency and scalability	The system shall support 500 concurrent users, 10,000 appointments per month, and 20 concurrent administrative users without degradation.

4. User Stories and Acceptance Criteria

Epic E1 — Public website and brand

US-01 — Brand home page

- **Role:** Visitor
- **Story:** As a visitor, I want to see the studio's brand and value proposition, so that I can decide whether to explore its services.
- **Priority:** Must · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** a visitor opens the website, **when** the home page loads, **then** the brand name, slogan, hero image, and booking call to action are visible.
  - **Given** the visitor uses a mobile device, **when** the page renders, **then** the layout adapts without horizontal scrolling.

US-02 — Service catalog

- **Role:** Visitor
- **Story:** As a visitor, I want to browse the service catalog with prices and durations, so that I can choose the service I need.
- **Priority:** Must · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** the visitor opens the catalog, **when** the page loads, **then** each service shows its name, description, duration, price, and assigned professionals.
  - **Given** the visitor selects a category, **when** the filter is applied, **then** only services in that category are displayed.

US-03 — Gallery and philosophy

- **Role:** Visitor
- **Story:** As a visitor, I want to see the studio's philosophy and gallery, so that I can trust the brand and its results.
- **Priority:** Must · **Estimate:** 2
- **Acceptance criteria:**
  - **Given** the visitor opens the gallery, **when** the images load, **then** each image has alternative text and belongs to an authorized client.
  - **Given** the visitor opens the philosophy page, **when** the content loads, **then** the brand story and values are displayed.

US-04 — Contact and location

- **Role:** Visitor
- **Story:** As a visitor, I want to contact the studio and find its location, so that I can ask questions or visit.
- **Priority:** Must · **Estimate:** 2
- **Acceptance criteria:**
  - **Given** the visitor is on the contact section, **when** they click the WhatsApp button, **then** a chat opens with the studio number 3132065167.
  - **Given** the visitor clicks the address, **when** the map opens, **then** it shows the studio location in Bogotá.

Epic E2 — Catalog and availability

US-05 — Filter by category and professional

- **Role:** Client
- **Story:** As a client, I want to filter services by category and professional, so that I can find the right option for me.
- **Priority:** Should · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** the catalog is loaded, **when** the client selects a professional, **then** only services offered by that professional are shown.

  - **Given** a professional is selected, **when** the client opens the profile, **then** the professional's name and specialties are displayed.

US-06 — Virtual gift card

- **Role:** Client
- **Story:** As a client, I want to buy and redeem a virtual gift card, so that I can give a service as a gift.
- **Priority:** Could · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** a client buys a gift card, **when** the purchase is completed, **then** a unique code is generated and sent by email.
  - **Given** a client redeems a valid code, **when** the code is applied to a booking, **then** the balance is discounted and the remaining balance is updated.

US-07 — Real-time availability

- **Role:** Client
- **Story:** As a client, I want to see real available time slots, so that I can choose a time that is actually free.
- **Priority:** Must · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the client selected a service and a professional, **when** the calendar loads, **then** only slots within working hours, without overlaps, and with the required anticipation are shown.
  - **Given** a slot is taken by another booking, **when** the calendar refreshes, **then** that slot is no longer selectable.

US-08 — Price and duration before booking

- **Role:** Client
- **Story:** As a client, I want to see the price and duration before booking, so that I can plan my visit and budget.
- **Priority:** Must · **Estimate:** 2
- **Acceptance criteria:**
  - **Given** the client selects a service, **when** the summary is displayed, **then** the price and duration are shown before confirmation.
  - **Given** the price changes, **when** the client reloads the page, **then** the updated price is displayed.

Epic E3 — Booking

US-09 — Start a booking

- **Role:** Client
- **Story:** As a client, I want to start a booking as a guest or registered user, so that I can reserve without unnecessary friction.

- **Priority:** Must · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** a visitor selects a service and a slot, **when** they continue, **then** the system asks for name, phone, and email, or allows login.
  - **Given** the client is registered, **when** they log in, **then** their data is pre-filled.

US-10 — Select date and time

- **Role:** Client
- **Story:** As a client, I want to select a date and time, so that I can book the appointment.
- **Priority:** Must · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the client is on the calendar, **when** they select a date, **then** the available slots for that date are displayed.
  - **Given** the client selects a slot, **when** they continue, **then** the slot is held temporarily for the configured window.

US-11 — Confirm the booking

- **Role:** Client
- **Story:** As a client, I want to confirm the booking and receive a summary, so that I know my appointment is registered.
- **Priority:** Must · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the client completed the form, **when** they confirm, **then** the appointment is created with status `pending` and a summary is displayed.
  - **Given** two clients try to confirm the same slot, **when** the second confirmation arrives, **then** the system rejects it and offers alternative slots.

US-12 — Confirmation and calendar file

- **Role:** Client
- **Story:** As a client, I want to receive a confirmation and add the appointment to my calendar, so that I do not forget it.
- **Priority:** Must · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** the appointment is registered, **when** the transaction completes, **then** the client receives an email with the appointment details.
  - **Given** the email is opened, **when** the client clicks the calendar link, **then** an ICS file is downloaded or the event is added to Google Calendar.

US-13 — Reschedule an appointment

- **Role:** Client
- **Story:** As a client, I want to reschedule my appointment within the allowed window, so that I can adapt to a change of plans.
- **Priority:** Should · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the appointment is more than the minimum notice away, **when** the client selects a new slot, **then** the appointment is moved and the old slot is released.
  - **Given** the appointment is inside the restricted window, **when** the client tries to reschedule, **then** the system explains the policy and offers contact with the studio.

US-14 — Cancel an appointment

- **Role:** Client
- **Story:** As a client, I want to cancel my appointment within the policy, so that the slot can be used by someone else.
- **Priority:** Should · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the appointment is outside the penalty window, **when** the client cancels, **then** the appointment becomes `cancelled` and the slot is released.
  - **Given** the appointment is inside the penalty window, **when** the client cancels, **then** the system shows the applicable policy before confirming.

Epic E4 — Client account and communications

US-15 — Register and log in

- **Role:** Client
- **Story:** As a client, I want to register and log in, so that I can manage my appointments and history.
- **Priority:** Should · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** a new client provides a valid email and password, **when** they submit the form, **then** the account is created and a verification email is sent.
  - **Given** a registered client enters valid credentials, **when** they log in, **then** they access their appointment dashboard.

US-16 — View appointments

- **Role:** Client
- **Story:** As a client, I want to see my upcoming and past appointments, so that I can track my visits.

- **Priority:** Should · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** the client is logged in, **when** they open "My appointments", **then** upcoming and past appointments are listed with date, service, professional, and status.
  - **Given** an appointment is upcoming, **when** the client opens it, **then** the reschedule and cancel options are shown according to policy.

US-17 — Read-only service history

- **Role:** Client
- **Story:** As a client, I want to view my service history in read-only mode, so that I can remember what was done.
- **Priority:** Should · **Estimate:** 2
- **Acceptance criteria:**
  - **Given** the client has completed services, **when** they open the history, **then** the services, dates, professionals, and notes marked as visible are displayed.
  - **Given** the client tries to edit a record, **when** they open it, **then** no edit action is available.

US-18 — Marketing consent

- **Role:** Client
- **Story:** As a client, I want to control my marketing consent, so that I only receive messages I authorized.
- **Priority:** Could · **Estimate:** 2
- **Acceptance criteria:**
  - **Given** the client registers, **when** they submit the form, **then** the consent checkbox is unchecked by default and the decision is stored.
  - **Given** the client withdraws consent, **when** the change is saved, **then** future promotional messages are blocked.

US-19 — Reminders and birthday benefit

- **Role:** Client
- **Story:** As a client, I want to receive reminders and a birthday benefit, so that I do not miss my appointment and feel valued.
- **Priority:** Could · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** an appointment is scheduled, **when** the reminder window is reached, **then** the client receives an email or WhatsApp reminder.
  - **Given** the client's birthday is within one week, **when** the birthday job runs, **then** a 10 % discount valid during the birthday week is issued.

Epic E5 — Agenda and CRM

US-20 — Daily and weekly agenda

- **Role:** Administrator
- **Story:** As an administrator, I want to see the daily and weekly agenda by professional, so that I can organize the operation.
- **Priority:** Should · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the administrator opens the agenda, **when** they select a date and professional, **then** the appointments are displayed in chronological order.
  - **Given** the administrator switches to week view, **when** the view loads, **then** all appointments for the week are visible by day and professional.

US-21 — Manage appointments

- **Role:** Administrator
- **Story:** As an administrator, I want to create, move, confirm, cancel, and mark no-show appointments, so that the agenda reflects reality.
- **Priority:** Should · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the administrator creates an appointment, **when** they save it, **then** the system validates availability and prevents overlaps.
  - **Given** an appointment is marked as no-show, **when** the status changes, **then** the event is recorded in the audit log and the slot is released.

US-22 — Client record and history

- **Role:** Administrator
- **Story:** As an administrator, I want to open a client record with contact data and service history, so that I can provide personalized service.
- **Priority:** Should · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** the administrator searches for a client, **when** they open the record, **then** contact data, appointments, services, and notes are displayed.
  - **Given** the client has no history, **when** the record opens, **then** an empty state with the create-appointment action is shown.

US-23 — Photos, formulas, and notes

- **Role:** Administrator / Professional
- **Story:** As a professional, I want to add photos, formulas, and notes to a service record, so that the next visit is consistent.
- **Priority:** Should · **Estimate:** 5
- **Acceptance criteria:**

  - **Given** a completed appointment, **when** the professional adds a photo, formula, or note, **then** the record is saved with the author and timestamp.
  - **Given** the professional does not have permission for a client, **when** they try to open the record, **then** access is denied.

US-24 — Duplicate client detection

- **Role:** Administrator
- **Story:** As an administrator, I want the system to detect possible duplicate clients, so that the history stays unified.
- **Priority:** Should · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** a new client is created with a phone or email that already exists, **when** the form is submitted, **then** the system shows a warning with the possible duplicate.
  - **Given** the administrator confirms the duplicate, **when** the records are merged, **then** the appointments and history are consolidated.

US-25 — Client segmentation

- **Role:** Administrator
- **Story:** As an administrator, I want to segment clients, so that I can send relevant messages and measure retention.
- **Priority:** Could · **Estimate:** 3
- **Acceptance criteria:**
  - **Given** the client database has activity, **when** the administrator opens segmentation, **then** frequent, inactive, birthday, high-value, and new clients are available as filters.
  - **Given** a segment is selected, **when** the administrator exports it, **then** a CSV with the selected clients is generated.

Epic E6 — Finance, inventory, and reports

US-26 — Daily cash closing

- **Role:** Administrator
- **Story:** As an administrator, I want to close the cash daily, so that income and expenses are under control.
- **Priority:** Could · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the day has transactions, **when** the administrator closes the cash, **then** income, expenses by category, and payment methods are summarized.
  - **Given** the counted cash differs from the system total, **when** the administrator saves the closing, **then** the difference is recorded with a justification.

US-27 — Professional commissions

- **Role:** Administrator
- **Story:** As an administrator, I want to calculate and settle professional commissions, so that payments are transparent.
- **Priority:** Could · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** a completed service, **when** the commission is calculated, **then** 55 % is applied by default and 60 % when the professional provides their own implements.
  - **Given** the week is closed, **when** the administrator settles commissions, **then** a per-professional summary is generated and marked as settled.

US-28 — Dashboard and monthly reports

- **Role:** Administrator
- **Story:** As an administrator, I want a daily dashboard and monthly reports, so that I can make decisions with data.
- **Priority:** Could · **Estimate:** 5
- **Acceptance criteria:**
  - **Given** the administrator opens the dashboard, **when** the page loads, **then** appointments, income, occupancy, and no-show figures for the day are displayed.
  - **Given** the administrator selects a month, **when** the report is generated, **then** sales by service and professional, average ticket, new vs. recurring clients, and retention are shown.

4. Story map

To summarize and present all of the user stories the following story map is presented

User story map — Beauty studio booking platform

28 stories · 4 roles · 6 epics · 107 story points

Role	E1 — Public website and brand	E2 — Catalog and availability	E3 — Booking	E4 — Client account and communications	E5 — Agenda and CRM	E6 — Finance, inventory, and reports
Visitor	**US-01 · Brand home page** **Must · 3 pts**					
	US-02 · Service catalog **Must · 3 pts**					
	US-03 · Gallery and philosophy **Must · 2 pts**					
	US-04 · Contact and location **Must · 2 pts**					
Client		US-05 · Filter by category and professional Should · 3 pts	**US-09 · Start a booking** **Must · 5 pts**	US-15 · Register and log in Should · 5 pts		
		US-06 · Virtual gift card *Could · 5 pts*	**US-10 · Select date and time** **Must · 5 pts**	US-16 · View appointments Should · 3 pts		
		US-07 · Real-time availability **Must · 5 pts**	**US-11 · Confirm the booking** **Must · 5 pts**	US-17 · Read-only service history Should · 2 pts		
		US-08 · Price and duration before booking **Must · 2 pts**	**US-12 · Confirmation and calendar file** **Must · 3 pts**	*US-18 · Marketing consent* *Could · 2 pts*		
			US-13 · Reschedule an appointment Should · 5 pts	*US-19 · Reminders and birthday benefit* *Could · 3 pts*		
			US-14 · Cancel an appointment Should · 5 pts			
Administrator					US-20 · Daily and weekly agenda Should · 5 pts	*US-26 · Daily cash closing* *Could · 5 pts*
					US-21 · Manage appointments Should · 5 pts	*US-27 · Professional commissions* *Could · 5 pts*
					US-22 · Client record and history Should · 3 pts	*US-28 · Dashboard and monthly reports* *Could · 5 pts*
					US-24 · Duplicate client detection Should · 3 pts	
					US-25 · Client segmentation *Could · 3 pts*	
Professional					US-23 · Photos, formulas, and notes Should · 5 pts	